

Design and performance analysis of a piezoelectric jetting dispensing valve

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Abstract

In order to meet the requirements of high frequency, high precision, and micro-scale dispensing in the field of microelectronics packaging, a piezoelectric jetting dispensing valve based on two-stage displacement amplifying mechanism was proposed. First, the overall structure and working principle of the proposed piezoelectric jetting valve were described. The displacement amplifying mechanism was designed, and the mathematical relationship between the output displacement and the structural parameters was established. In addition, the performance and the structural size of the displacement amplifying mechanism were analyzed and optimized by using the finite element analysis software. The influences of driving pressure, glue viscosity, needle displacement, and other parameters on droplet diameter and jetting velocity in the process of dispensing were simulated and analyzed. In order to obtain the optimal nozzle structure, the changing curves of glue flowing velocity with different nozzle seal forms, nozzle cone angles, and outlet inner diameters were compared and analyzed. Finally, the prototype of the piezoelectric jetting dispensing valve was machined, and experimental study was performed. The influence laws of driving pressure, glue viscosity, driving voltage, and other parameters on the diameter of droplet were explored. The performance of the prototype was evaluated from three aspects of dispensing frequency, consistency, and minimum dispensing droplet diameter. Results show that the highest dispensing frequency is 230 Hz, the deviation of dispensing consistency is $\pm 8.77\%$, and the minimum dispensing droplet diameter is 0.54 mm. The experimental results verify the high frequency, consistency, and micro-scale dispensing performance of the piezoelectric jetting dispensing valve based on the displacement amplifying mechanism, which provides a reference for the research of piezoelectric high-frequency jetting dispensing.

Keywords

Piezoelectric jetting dispensing valve, displacement amplifying mechanism, dispensing performance, simulation analysis, experimental study

1. Introduction

With the rapid development of modern science and industry technology, the information technology industry has higher requirements for integrated circuits and related dedicated equipment. Due to the pursuit of thin, short, and integrated electronic products nowadays, it is crucial to improve microelectronics bonding, 3D micro-assembly, high-density packaging, and other technologies. Fluid dispensing technology is widely used in the field of microelectronics packaging (Sun et al., 2011; Zhou et al., 2019), such as chip packaging, fixation, bottom filling, and other processes (Babiarz et al., 1997). Microelectronic packaging has strict requirements on fluid dispensing technology, such as high frequency, high precision, micro-scale dispensing, and consistency of glue droplets (Deng et al., 2018).

According to different working principles, fluid dispensing technology is mainly divided into contact

dispensing and non-contact dispensing. The contact dispensing has high requirements for the height positioning accuracy, which requires the dispensing valve to move repeatedly in the direction of the Z axis, so it is easy to collide and cause damage. In addition, glue with high viscosity is easy to appear tailing phenomenon in the process of dispensing glue, which is difficult to achieve the consistency of glue droplets, and does not meet the needs of high-frequency, micro-scale, and stable dispensing (Wang et al., 2013). Different from contact dispensing, non-contact dispensing technology is more suitable for the development of today's

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microelectronics packaging, its high frequency, high precision, high reliability, and micro-scale dispensing characteristics are more in line with the needs of future development. Jetting dispensing is the most important form of non-contact dispensing technology. According to different driving modes, it can be divided into pneumatic type (Amirzadeh Goghari and Chandra, 2007; Cheng et al., 2005), thermal bubble type (Ko et al., 2021) and piezoelectric type, etc.

Compared with other jetting dispensing technologies, piezoelectric dispensing has attracted the attention of scholars and engineers due to its high frequency, low noise, stability, and strong glue applicability (Li et al., 2017). Ashley and Adamson (2008) combined piezoelectric technology with a new valve design to develop a new high-speed, small-dot solder paste dispensing technology. Fu et al. (2015) proposed a new type of hydraulic amplifying piezoelectric jetting dispensing machine, and established a closed-loop control system using PID algorithm, which improved the performance of piezoelectric jetting dispensing system, obtaining stable spraying droplets with dispensing speed of 0.023 L/min and droplet diameter of 0.8 mm. Luo et al. (2020) improved the stability and life of piezoelectric dispensers by studying the damage mechanism of piezoelectric ceramics in the process of circulating dispensing, but the tested highest frequency of the jetting dispenser was 65 Hz, with the smallest droplet diameter of 1.07 mm. Nguyen et al. (2007, 2008) set up the mathematical model of a new type of piezostack-driven jetting dispenser by using lumped parameter method, and used hydraulic amplification and resonance principles to amplify the displacement of the piezoelectric stack and drive the needle to achieve the injection of the glue, achieving the maximum working frequency around 700 Hz and the maximum dispensing flow rate of 225 mg/s.

Because the displacement generated by the piezoelectric actuator is too small to drive the needle directly, a micro displacement amplifying mechanism is usually configured to amplify the displacement generated by the piezoelectric actuator and then transfer it to the needle for work. There are many types of displacement amplifying mechanism, such as lever type, triangle type, and diamond type. Zhou et al. (2010) designed a triple micro-displacement amplifying mechanism according to the principle of lever. Zheng et al. (2020) designed a rhomboid micro-displacement amplifier to drive piezoelectric dispensing equipment, which can achieve a displacement of 0.3–0.5 mm. Yang et al. (2007) designed a triangular flexural hinge amplifier that could increase the output displacement by about five times.

In the process of installing electronic components such as inductors and capacitors to the PCB board, there will be a glue process, which requires the diameter of the glue drop to be about 1 mm, and requires high frequency dispensing to improve efficiency. At present,

the existing dispensing valves at home and abroad are expensive and patented. In this paper, a piezoelectric jetting dispensing valve based on two-stage displacement amplifying flexible hinge mechanism was proposed to be applied in the above scenarios. The displacement amplifying mechanism was designed and optimized by the modeling software and finite element analysis software. Then, the changing curves of glue flowing velocity under different nozzle structure parameters were compared to optimize the design of the nozzle structure. Next, the effects of the parameters on the diameter of droplets and the velocity of jetting were analyzed by simulation, which provided a theoretical basis for the experiment. Finally, through the experiment, the influence law of driving pressure, glue viscosity, driving voltage, and other parameters on the diameter of the glue droplets was explored, and the influence of dispensing frequency, consistency, and minimum glue droplet on the performance of dispensing was studied.

2. Structural design of piezoelectric jetting dispensing valve

2.1. Overall structure and working principle of piezoelectric dispensing valve

Figure 1 shows the structure diagram of the piezoelectric jetting dispensing valve. As the core power source, the piezoelectric actuator outputs force and displacement under the action of the driving power. Preloading bolts are used to apply prestress to the piezoelectric actuator. The preloading bolt exerts a pretightening force on the piezoelectric stack through the spacer to play the role of fixing. The degree of pretightening is limited by not causing the lower half of the amplifier lever to tilt. The displacement amplifying mechanism adopts a flexible two-stage amplifying method to amplify the output displacement of the piezoelectric

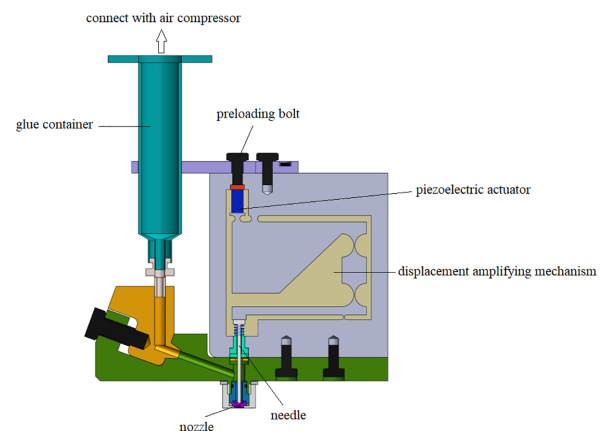


Figure 1. Structure diagram of piezoelectric jetting dispensing valve.

actuator and transmit it to the needle. The separation and closure of the needle and the nozzle complete the jetting of the glue droplets. In addition, the glue container is connected with an air compressor, so that glue is in a stable pressure state during the flow, which is conducive to the dispensing of glue.

The working principle of the piezoelectric jetting dispensing valve is as follows: As shown in Figure 2(a), when no voltage is applied, the piezoelectric actuator, the displacement amplifying mechanism, the needle, and the back spring are tightly connected under the preloading force of the preloading bolt. The needle holds its initial position under the action of the back spring. Under the action of the surface tension of the glue and the viscous force between the glue and the nozzle wall, the glue will not flow out of the nozzle, which can be confirmed by the test. When a positive square wave voltage is applied to the piezoelectric actuator, the actuator is extended. The displacement amplifying mechanism pushes the needle to move down and compresses the back spring. At this time, the velocity of the glue liquid increases and the glue ejects from the nozzle, as shown in Figure 2(b). When the voltage drops to zero, the piezoelectric actuator retracts, and the needle moves upward to return to the state of no voltage under the action of the back spring.

Under the action of alternating high frequency square wave voltage, the needle vibrates at a high frequency, so as to achieve the effect of ejecting at a high frequency. Overcoming viscous force and surface tension is a prerequisite for fluid ejection, which is related to the dimensionless numbers, Weber number W_e and Reynolds number R_e (Huang et al., 2020), and the following conditions must be satisfied:

$$W_e = \frac{\rho U^2 d}{\sigma} > 1 \quad (1)$$

$$R_e = \frac{\rho U d}{\mu} > 1 \quad (2)$$

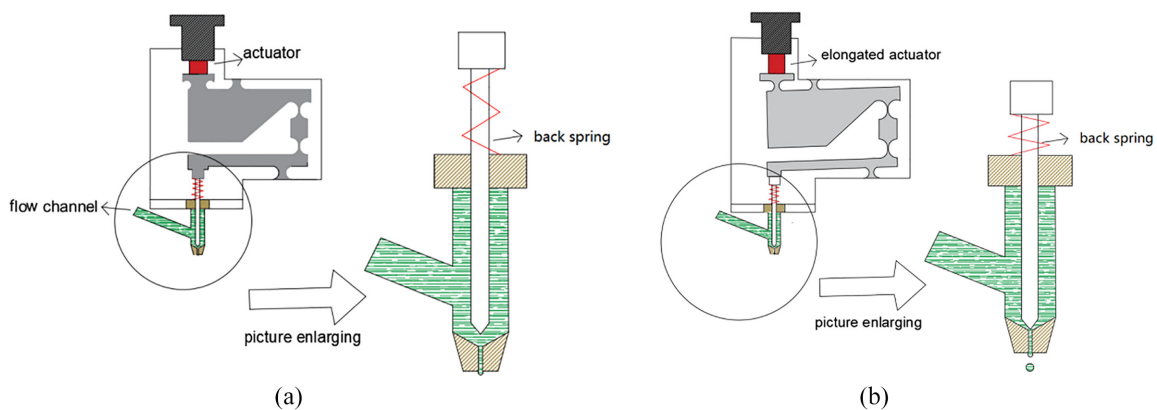


Figure 2. State of the device: (a) no voltage is applied and (b) voltage is applied and the actuator is extended.

In the formulas, σ is the surface tension coefficient, μ is the hydrodynamic viscosity, ρ is the fluid density, U is the fluid velocity, and d is the nozzle aperture. The velocity U can be idealized as the average velocity when the needle moves downward.

2.2. Design of displacement amplifying mechanism

A new type of flexible two-stage displacement amplifying mechanism is proposed, as shown in Figure 3(a), and its rigid structure is shown in Figure 3(b). A and D are fixed hinges, the piezoelectric actuator is placed at O, rod OAB is the first-stage amplifying mechanism, and rod CDE is the second-stage amplifying mechanism. The two are connected through rod BC, and the input displacement and output displacement directions are both vertical. Since the maximum output displacement of the piezoelectric actuator is only about 10 μm , and the output displacement of the needle is generally more than 200 μm in practical application (Zhou et al., 2018b), the amplifying factor of the two-stage displacement amplifying mechanism designed in this paper should be more than 20 times.

2.2.1. Static calculation analysis of displacement amplifying mechanism. The design model of the two-stage displacement amplifying mechanism is shown in Figure 4. U_{in} and U_{out} represent the input and output displacements of the mechanism respectively. J_1, J_2, J_3, J_4 , and J_5 are the flexible hinges from the input end to the output end in turn. l_1 and l_2 represent the distance from the input end of the first-stage lever to the hinge J_2 and the distance from the output end to the hinge J_2 respectively. l_3 and l_4 represent the distance from the input end of the second-level lever to the hinge J_5 and the distance from the output end to the hinge J_5 , respectively.

For the arc-shaped flexible hinge used in this paper, many scholars have proposed accurate and concise theoretical calculation formulas (Acer and Sabanovic, 2011; Bu et al., 2018). In summary, the relationships

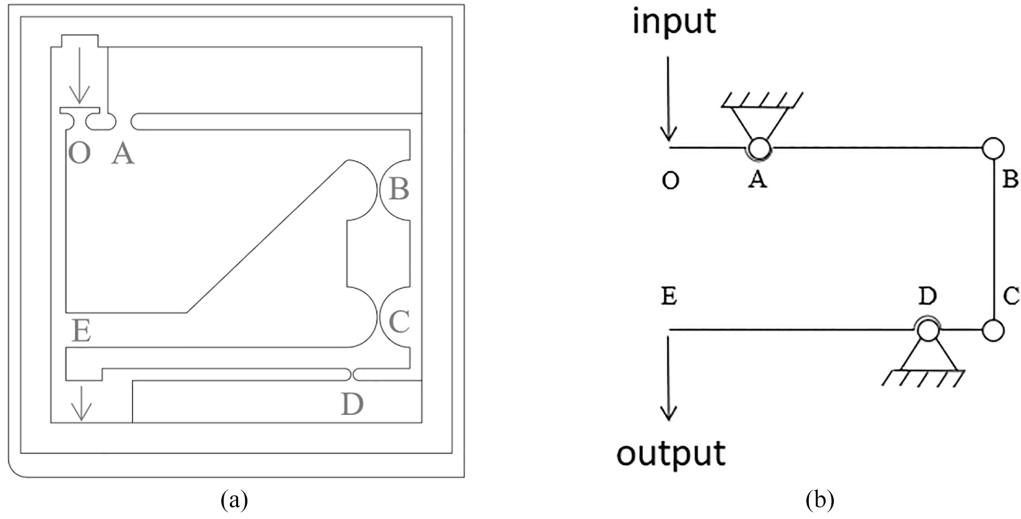


Figure 3. Two-stage displacement amplifying mechanism: (a) flexible mechanism model and (b) diagram of rigid structure.

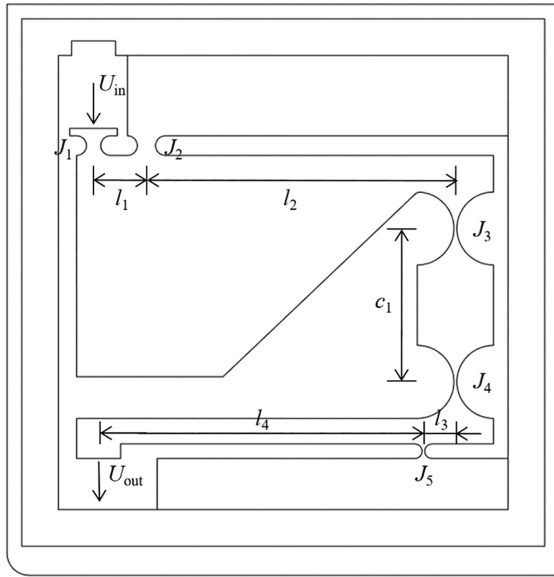


Figure 4. Model of two-stage displacement amplifying mechanism.

between torque M_i and angle α_i , axial force F_i and axial deformation Δ_i of arc-shaped flexible hinge are as follows:

$$\begin{aligned} M_i &= K_\alpha \cdot \alpha_i \\ F_i &= K_x \cdot \Delta_i \end{aligned} \quad (3)$$

where K_α and K_x represent the rotational stiffness and tensile stiffness of the flexible hinge, respectively. $i = 1, 2, 3, 4$, and 5 correspond to the five flexible hinges in the Figure 5, respectively.

The angles α at the five flexible hinges have the following relationships with the angles θ of the two levers:

$$\alpha_1 = \alpha_2 = \theta_1$$

$$\alpha_3 = \alpha_4 = \theta_2 - \theta_1$$

$$\alpha_5 = \theta_2 \quad (4)$$

The force analysis of the two levers was carried out respectively, and the relationship between the torque, force, and displacement is shown in Figure 5. From Figure 5(a) the force balance can be obtained:

$$\begin{aligned} F_1 + F_3 &= F_2 \\ M_1 + M_2 + F_3 \cdot l_2 &= M_3 + F_1 \cdot l_1 \end{aligned} \quad (5)$$

For the first-stage lever, the theoretical input displacement U_{in} of the displacement amplifying mechanism will be lost due to the compression deformation generated by the hinge J_1 , so its actual input displacement is:

$$U_1 = U_{in} - \Delta_1 \quad (6)$$

The actual output displacement of the first-stage lever can be obtained as follows:

$$U_3 = \theta_1 \cdot l_2 - \Delta_2 \quad (7)$$

The angle θ_1 of the lever can be expressed as (valid only under the condition: $\theta_1 \ll 1$):

$$\theta_1 = \frac{U_1 - \Delta_2}{l_1} = \frac{U_{in} - \Delta_1 - \Delta_2}{l_1} \quad (8)$$

Similarly, for the second-stage lever, it can be obtained from the force balance in Figure 5(b):

$$\begin{aligned} F_4 &= F_5 = F_3 \\ F_4 \cdot l_3 + M_5 &= M_4 \end{aligned} \quad (9)$$

The actual input displacement of the second-stage lever is:

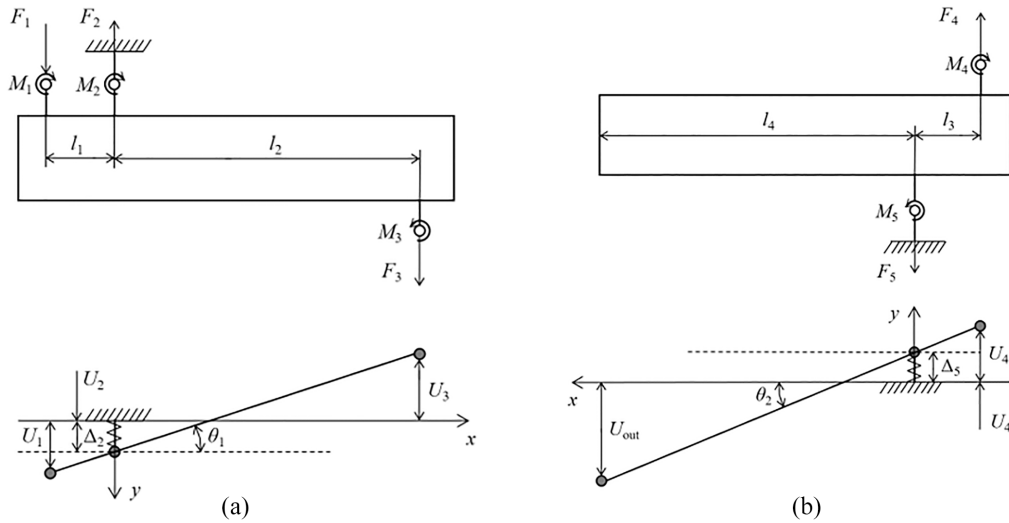


Figure 5. Force analysis of two-stage displacement amplifying mechanism: (a) diagram of the first-stage lever and (b) diagram of the second-stage lever.

$$U_4' = U_4 - \Delta_4 = [1 + \sin(\theta_2 - \theta_1)]U_3 - \Delta_4 \quad (10)$$

The actual output displacement of the second-stage lever can be obtained as follows:

$$U_{out} = \theta_2 \cdot l_4 - \Delta_5 \quad (11)$$

The angle θ_2 can be expressed as:

$$\theta_2 = \frac{U_4' - \Delta_5}{l_3} = \frac{[1 + \sin(\theta_2 - \theta_1)]U_3 - \Delta_4 - \Delta_5}{l_3} \quad (12)$$

Through the calculation and analysis of the statics model of the two-stage displacement amplifying mechanism, the relationship between the actual output displacement and each parameter can be obtained, which establishes the theoretical basis for simulation analysis and experimental research.

2.2.2. Simulation analysis and optimization of displacement amplifying mechanism. For simulation analysis, the maximum output displacement of the piezoelectric actuator was set to 10 μm . The dimensions of each part of the amplifying mechanism were calculated and designed by the above output displacement calculation formula, and the three-dimensional model of the amplifying mechanism was established. Then the displacement amplifying mechanism was simulated and analyzed. In the simulation, spring steel was selected as the structural material, and the density, elastic modulus, and Poisson's ratio were set as 7820 kg/m^3 , 197 GPa, and 0.282, respectively. After grid division, a fixed constraint was applied outside the mechanism and an input displacement of 10 μm was applied to the mechanism. Then the simulation diagrams of stress, strain, and

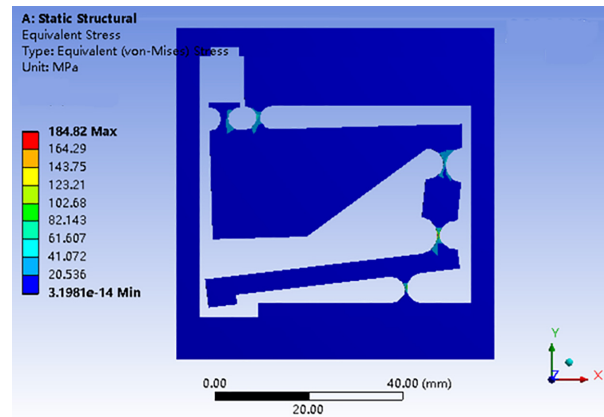


Figure 6. Stress diagram of displacement amplifying mechanism.

total displacement were obtained, as shown in Figures 6 to 8.

It can be seen from Figures 6 and 7 that the maximum stress and strain of the displacement amplifying mechanism both appear in the middle of flexible hinges, in which the maximum stress is about 185 MPa, far less than the yield strength of spring steel of 800 MPa, which is in line with the actual application requirements. Figure 8 shows that the maximum displacement deformation is on the contact surface between the amplifying mechanism and the needle, and the maximum output displacement is about 147 μm , with the magnification less than 15 times. Therefore, the initial model still needs further size optimization.

Because the structure and spacing of each flexible hinge of the mechanism are different and there are many dimensions involved, the size optimization was carried out in two stages in order to reduce the

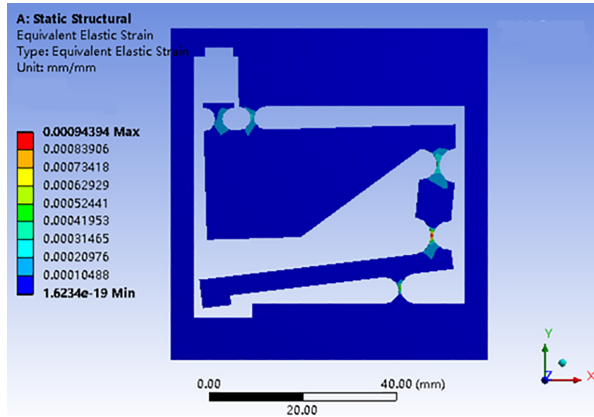


Figure 7. Strain diagram of displacement amplifying mechanism.

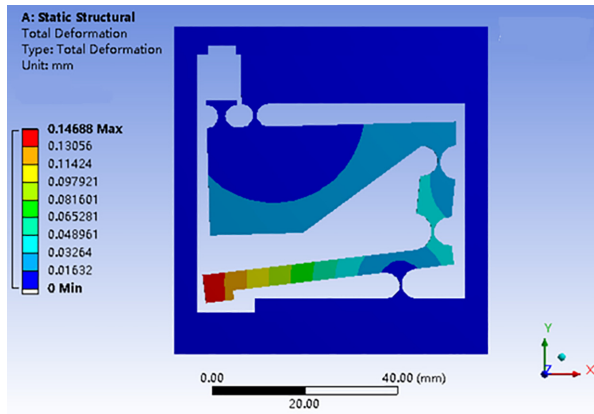


Figure 8. Displacement diagram of displacement amplifying mechanism.

calculation points of the optimization design modeling. According to formulas (11) and (12), the hinge deformation and hinge spacing are the main factors affecting the output displacement. Therefore, in the first stage, the arc radius and the middle thickness of each flexible hinge were optimized, and in the second stage, the spacing between each flexible hinge was optimized.

In the software Solidworks, the size to be optimized in the first stage was parameterized. The radius of the arc at O and A of the flexible hinges in Figure 9 is defined as R_1 , the radius of the arc at B and C is R_2 , and the radius of the arc at D is R_3 . The thickness of the middle part at O is b_1 , the thickness of the middle part at A is b_2 , the thickness of the middle part at B and C is b_3 , and the thickness of the middle part at D is b_4 . The parameterized model was imported into the software ANSYS Workbench, and the Response Surface optimization under the Design Exploration module was used as the optimization algorithm. R_1 , R_2 , R_3 , b_1 , b_2 , b_3 , and b_4 were used as the input parameters. Total Deformation Maximum of the mechanism was used as the output parameter to solve, and the influence laws of

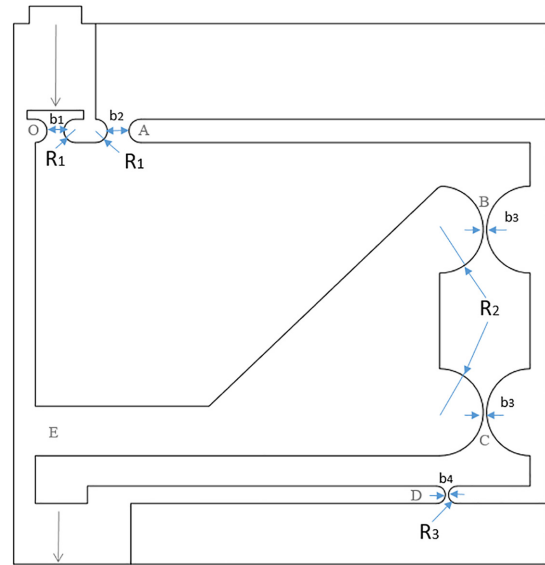


Figure 9. Optimized parameter diagram.

the optimization parameter variables R_1 , R_2 , R_3 , b_1 , b_2 , b_3 , and b_4 on the output displacement were obtained, as shown in Figure 10.

The sensitivity coefficient of each parameter can be obtained by analyzing the relationship between single parameter and output displacement. As shown in Figure 11, b_3 has a much greater impact on the output displacement of the second-stage displacement amplifying mechanism than other parameters. b_4 and R_1 have the second effect. R_2 , R_3 , b_1 , and b_2 are the least sensitive to the output displacement. Figure 12 shows the 3D response diagram of the vertical output displacement and two of the three parameters b_3 , b_4 , and R_1 , whose sensitivity coefficient are the largest.

According to the solution results obtained in the Design Exploration module, three groups of combined points with maximum output displacement can be obtained, as shown in Table 1. Since parameters b_3 and b_4 are highly sensitive and meet the actual application requirements, the displacement magnification of the recommended point 1 is far better than that of the other points. Finally, $R_1 = 1.35$ mm, $R_2 = 5$ mm, $R_3 = 1$ mm, $b_1 = 1.94$ mm, $b_2 = 2.5$ mm, $b_3 = 0.4$ mm, and $b_4 = 0.35$ mm after the first stage optimization.

In the first stage optimization, the relevant size of flexible hinges itself was mainly optimized. On this basis, the parameter variables for the second stage optimization are X_1 , X_2 , and L_1 to L_6 , as shown in Figure 13, which mainly include the spacing between the flexible hinges and the cutting clearance of the valve body.

The sensitivity coefficient of each parameter can be obtained by comprehensively analyzing the influence of the eight parameter variables on the output

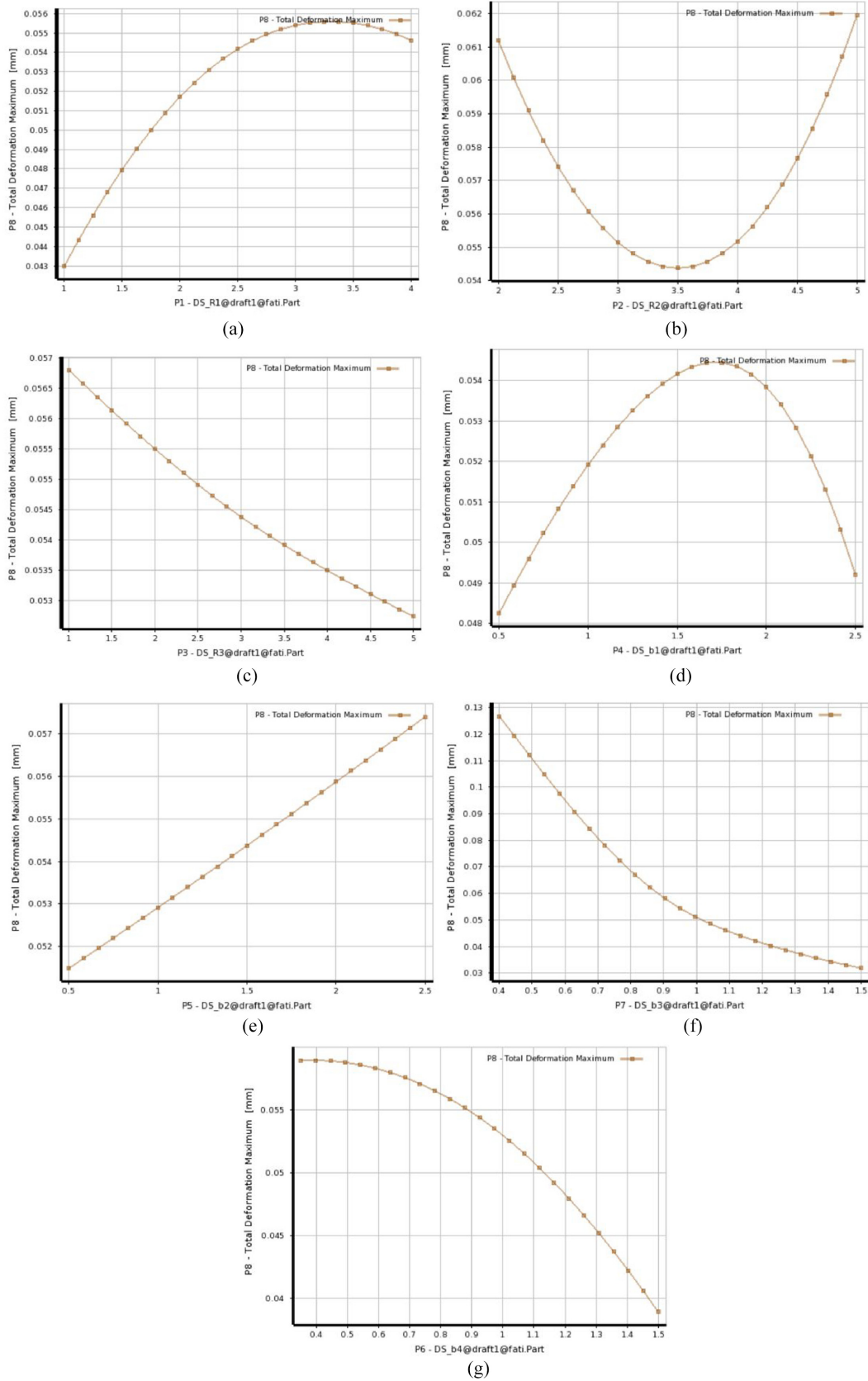


Figure 10. Relation curve between first-stage optimization parameters and output displacement: (a) effect of R_1 on output displacement, (b) effect of R_2 on output displacement, (c) effect of R_3 on output displacement, (d) effect of b_1 on output displacement, (e) effect of b_2 on output displacement, (f) effect of b_3 on output displacement, and (g) effect of b_4 on output displacement.

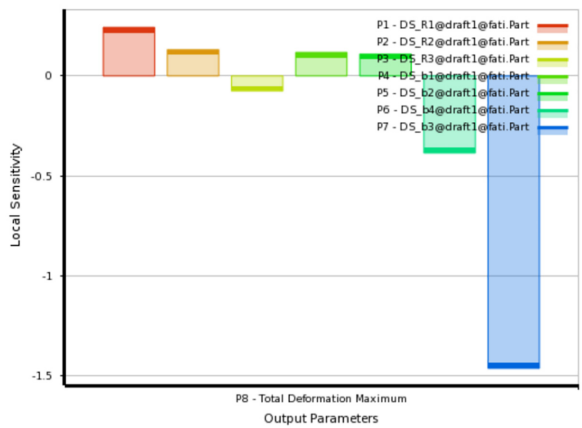


Figure 11. Sensitivity coefficient of each parameter in the first stage to the output displacement.

displacement, as shown in Figure 14. Among them, the highest sensitivity is L_6 , L_2 , L_4 , and L_5 , and the other four parameters have lower sensitivity. After processing the results, the 3D diagram of the response of L_6 , L_2 ,

and L_4 to the vertical output displacement when changing within the set optimization value range can be obtained, as shown in Figure 15. According to the solution results in the Design Exploration module, the recommended combined point with the largest output displacement can be obtained. Taking L_6 with the largest sensitivity as the primary design parameter, combining with the actual size application requirements, it was obtained that $L_1 = 2.3$ mm, $L_2 = 44.8$ mm, $L_3 = 5$ mm, $L_4 = 11$ mm, $L_5 = 3.5$ mm, $L_6 = 10.4$ mm, $X_1 = 5.7$ mm, and $X_2 = 20$ mm. The optimized size was used to re-model the two-stage displacement amplifying mechanism and the modal analysis was performed. The displacement cloud diagram is shown in Figure 16. The maximum output displacement is about 244 μm , and the magnification factor is 24 times, which meets the design requirements. The amplifying mechanism model is established for subsequent simulation and experiments.

In order to verify the output displacement simulation results of the displacement amplifying mechanism,

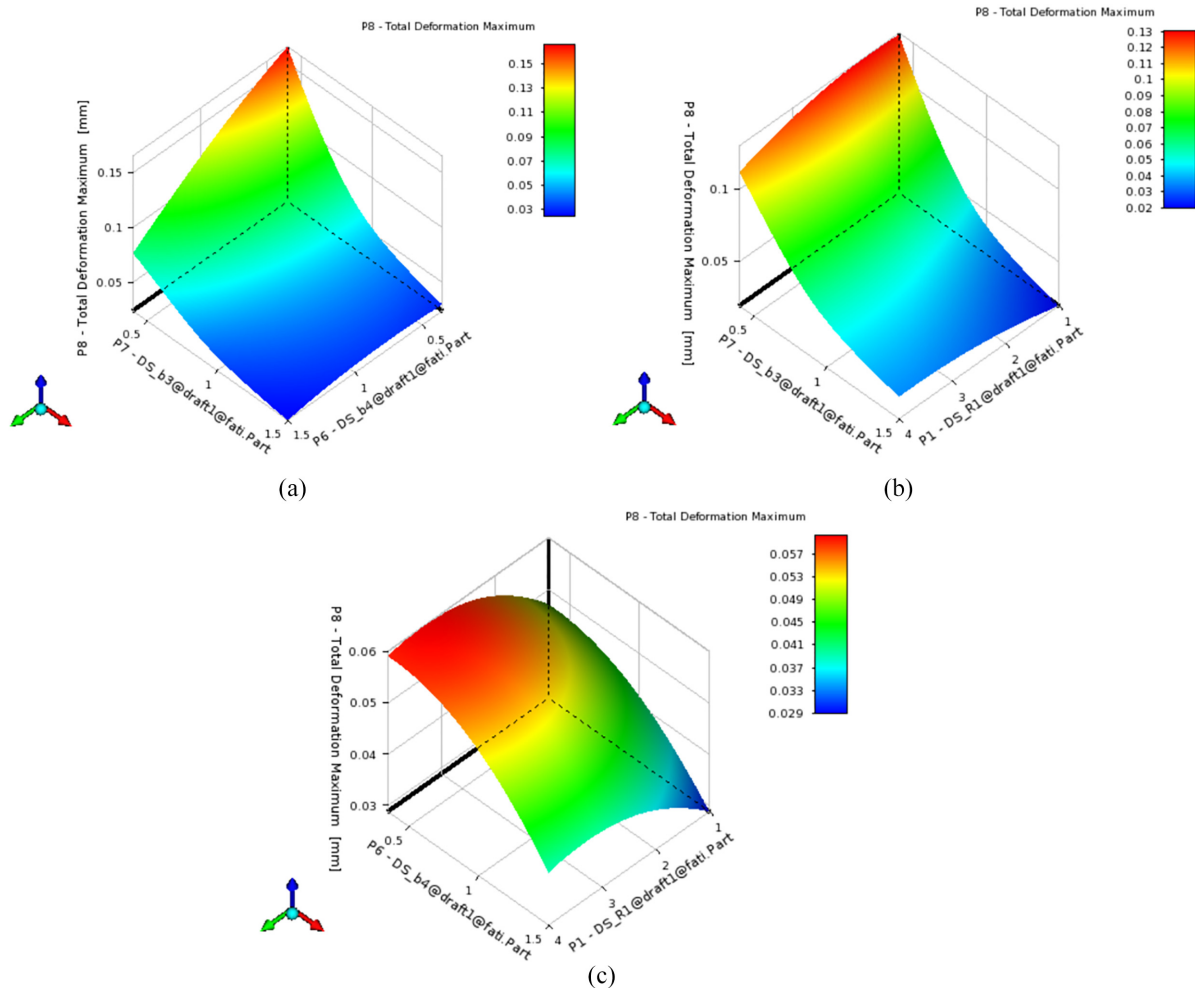
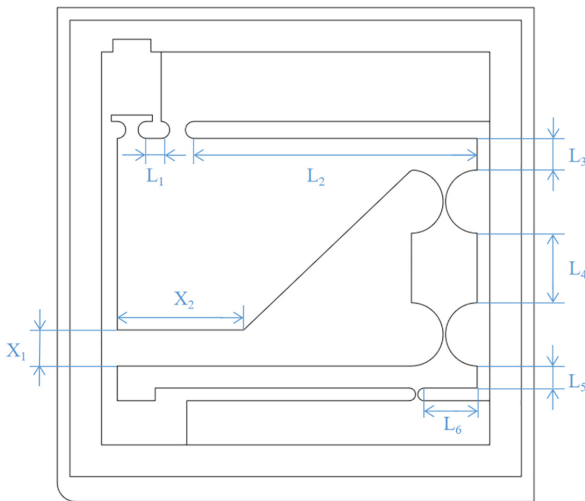
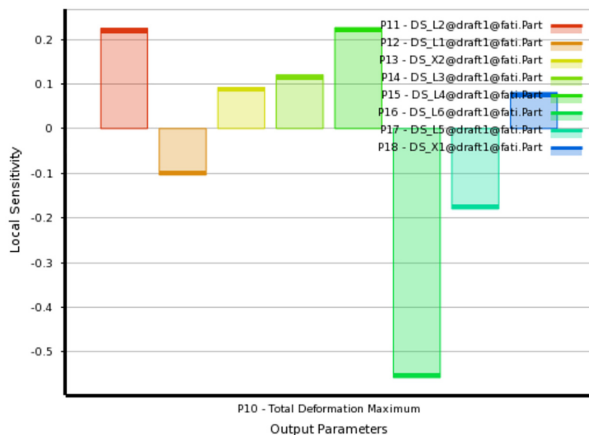


Figure 12. Relation surface diagram of b_3 , b_4 , R_1 , and output displacement: (a) b_3 , b_4 , and output displacement response surfaces, (b) b_3 , R_1 , and output displacement response surfaces, and (c) b_4 , R_1 , and output displacement response surfaces.

Table 1. Recommended combined points optimized based on output displacement.

	Candidate point 1	Candidate point 2	Candidate point 3
R1 (mm)	1.3464	1.0015	2.5225
R2 (mm)	5	2.0015	4.6206
R3 (mm)	1	1.002	1.7537
b1 (mm)	1.9417	0.501	1.3938
b2 (mm)	2.5	0.501	1.4581
b3 (mm)	0.4	0.40055	0.40205
b4 (mm)	0.35	0.35058	0.47758
Total deformation maximum (mm)	0.21544	0.18464	0.17764

**Figure 13.** Optimization parameters in the second stage optimization.**Figure 14.** Sensitivity coefficients of parameters to output displacements in the second stage optimization.

a laser vibration measurement experiment was carried out. The output displacement of the piezoelectric stack is about 10 μm at 100 V, which is consistent with the simulation input condition. As shown in Figure 17, under the periodic square wave excitation of 100 Hz

and 100 V, the maximum amplitude of the end of the displacement amplifying mechanism of the dispensing valve was measured by the laser vibrometer (Polytec VibroFlex Connect VFX-F-110). As shown in Figure 18, the peak amplitude is 198.6 μm , which is close to but smaller than the simulation result of 244 μm .

The working frequency of the dispensing valve should be less than its natural frequency to prevent the structure from being damaged due to resonance. Through modal analysis, the first-order resonant frequency of the amplifying mechanism is 436 Hz, as shown in Figure 19.

3. Simulation analysis of dispensing process

In the dispensing process, the flow velocity of glue and the size of glue droplets are the two main parameters to measure the dispensing quality. The faster flow velocity of droplets can attach more precisely, and smaller droplets can meet the requirements of micro-scale dispensing (Huang et al., 2019). The dispensing process of the piezoelectric jetting valve is essentially a process in which the flow state of the glue changes under the action of the needle and nozzle. The structure of the nozzle, the viscosity of glue, the stroke of the needle, and the pressure of the flow channel will directly affect glue droplets. In this chapter, the process was simulated and analyzed by using software COMSOL. The simulation condition is 0.2 mm displacement of the needle output, and the maximum flow velocity at the front of the nozzle was taken as the reference velocity. Then the influence of each factor was discussed, which provides theoretical guidance for the subsequent experimental study.

3.1. Influence of nozzle structure on dispensing performance

The sealing form, cone angle, outlet diameter, and other structural parameters of the nozzle have an influence on the dispensing performance of the dispensing valve (Lu et al., 2015; Zhou et al., 2018a). Therefore, it is

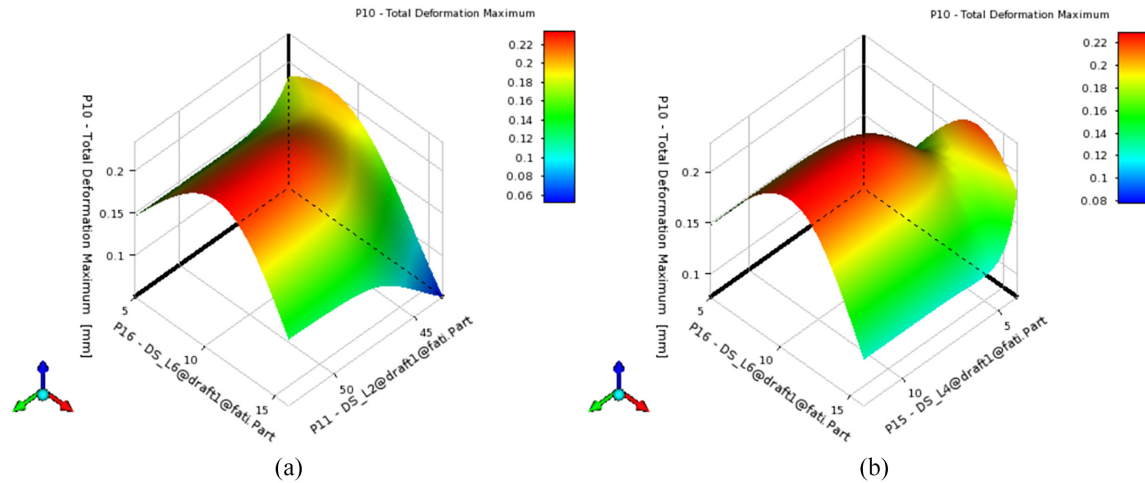


Figure 15. Relation surface diagram of L_6 , L_4 , L_2 , and output displacement: (a) L_6 , L_2 , and output displacement response surfaces and (b) L_6 , L_4 , and output displacement response surfaces.

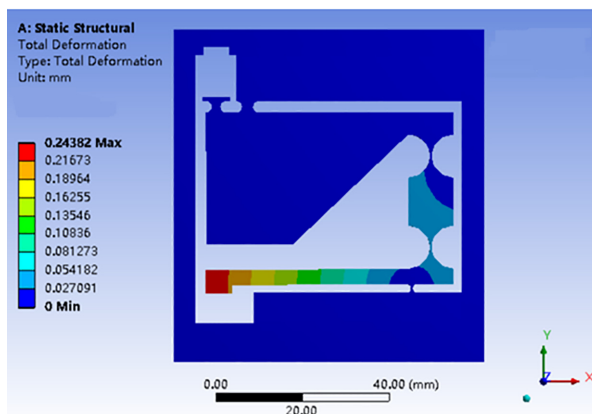


Figure 16. Displacement cloud diagram of optimized displacement amplifying mechanism.

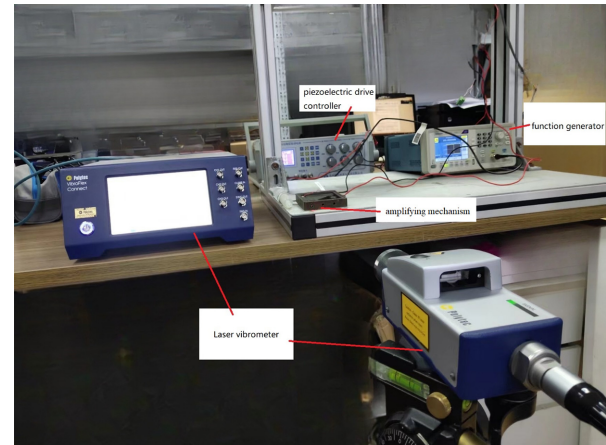


Figure 17. Measurement of the output displacement of the amplifying mechanism by using laser vibrometer.

necessary to study the influence law, so as to provide theoretical support for the nozzle design.

3.1.1. Sealing of needle and nozzle. The sealing effect of the needle and the nozzle plays a role in shutting off the flow of the glue, so that the dispensing valve can dispense glue at a certain frequency. The sealing of the needle and nozzle is mainly divided into three mating modes: cone-cone, cone-sphere, and sphere-sphere (Wang et al., 2014), as shown in Figure 20.

Taking the sealing performance as the analysis factor, the cone-cone, and sphere-sphere contact are better, which are surface contact type. However, the processing difficulty of needles and nozzles in the sphere-sphere mating mode is much greater than that in the cone-cone mating mode. Therefore, the cone-cone mating mode was selected for sealing of the needle and nozzle.

3.1.2. Angle of nozzle cone. The fluid region in the nozzle was taken as the simulation model, and with the same other parameters, the glue flow velocity simulation with cone angle between 60° and 120° was performed. The velocity cloud diagrams of some angles are shown in Figure 21, and the relation curve between velocity and angle is shown in Figure 22.

It can be seen from Figure 22 that the flow velocity of the glue decreases monotonically with the increase of the cone angle, but the range of change is small and concentrated in 8.1–9.3 m/s. This is mainly caused by the local resistance caused by the sudden change of the flow direction of the glue or sudden change of the shape of the channel (Chorin and Marsden, 1979). When the cone angle is 120° , the vertical downward flow turning angle of the glue is the largest, and the local resistance is the largest. Therefore, a smaller cone angle should be selected in an appropriate range to facilitate the

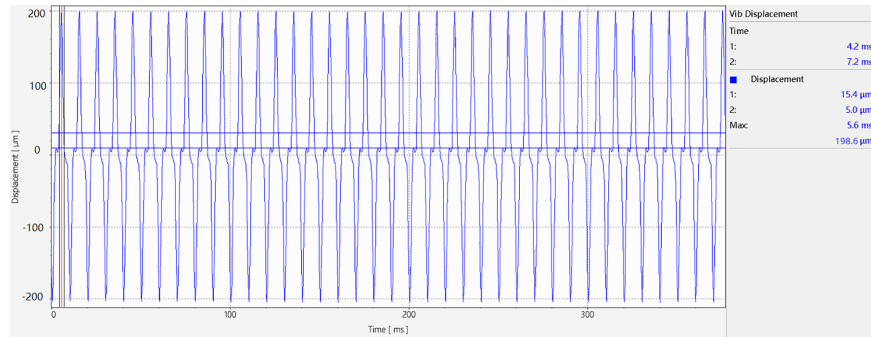


Figure 18. Experimental measuring results of the output displacement.

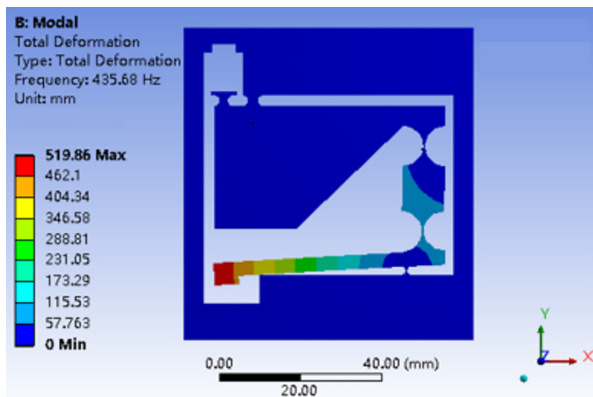


Figure 19. First-order vibration shape diagram of the displacement magnifying mechanism.

dispensing of glue droplets while avoiding the influence of too small cone angle on the shape of the glue droplets.

3.1.3. Diameter of nozzle outlet. With the same parameters, including the cone angles of 120° , the flow velocity of glue with the inner diameter of 0.15–0.45 mm was simulated. The flow velocity cloud diagrams are shown in Figure 23, and the relation curve between the flow velocity and the inner diameter of the outlet is shown in Figure 24.

It can be seen from Figure 24 that when the inner diameter of outlet is 0.15–0.2 mm, the flow velocity of the glue increases with the increase of the outlet size, and the maximum flow velocity is 8.15 m/s at 0.2 mm. When the inner diameter is 0.2–0.45 mm, the flow velocity decreases rapidly at first, and then fluctuates stably around 6 m/s. The reason is that when the inner diameter of the outlet is small, the viscous resistance effect of the glue is obvious. And if the diameter is too small, it is difficult to eject and even block the nozzle. When the diameter is too large, the extrusion action of the needle is difficult to be transmitted to the outlet, resulting in a decrease in the flow velocity. Therefore, on the premise of avoiding the difficulty of ejecting glue caused

by the small inner diameter of the outlet, a smaller inner diameter size should be selected in an appropriate range to obtain a larger ejection speed.

3.2. Influence of driving pressure on dispensing performance

The cone-cone nozzle structure with 60° cone angle and 0.2 mm outlet inner diameter obtained from the above simulation analysis was adopted to simulate and analyze the influence law of driving air pressure on dispensing performance. In addition, the viscosity of the glue was preset to 1000 cps, the displacement of the needle was preset to $150\ \mu\text{m}$, and the vibration frequency of the needle was preset to 100 Hz.

When pressure P_1 of 0.8 MPa was applied at the glue inlet and gauge pressure P_0 was defined as 0 MPa at the outlet, the dispensing performance distribution cloud diagram was obtained by simulation as shown in Figure 25. The maximum velocity of the glue is at the nozzle outlet which is 6.83 m/s, and its velocity distribution is shown in Figure 25(a). When the glue is ejected into the air from the nozzle, the diameter of the droplet is 0.84 mm, and its volume fraction distribution is shown in Figure 25(b), where the volume fraction 0 and 1 indicates that there is no glue in the unit volume (completely air) and the unit volume is full of glue, respectively.

In order to explore the influence of driving pressure on dispensing performance, under the premise of keeping other conditions unchanged, the maximum flow velocity of the glue and the diameter of the glue droplet were solved when P_1 was 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, and 0.8 MPa. And the result is plotted the changing curve as shown in Figure 26.

It can be seen from Figure 26 that both the maximum flow velocity of the glue and the diameter of the glue droplet increase with the increase of the driving pressure, and the slope of the change of the flow velocity is basically the same. The reason is that when the driving pressure increases, the initial velocity at the fluid inlet increases, so that the velocity at the nozzle

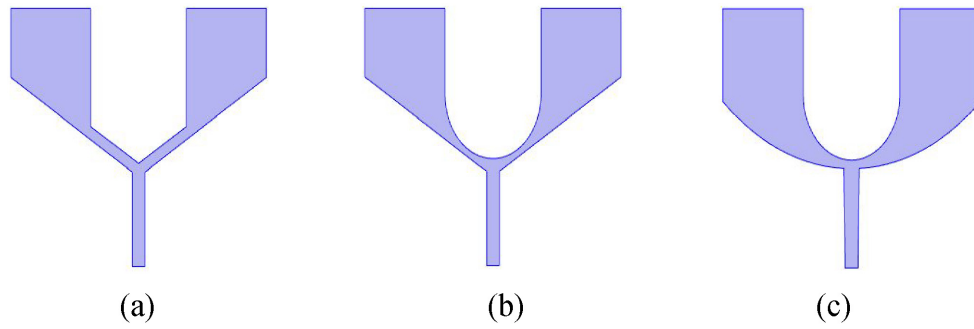


Figure 20. Sealing form of the needle and nozzle: (a) cone-cone, (b) cone-sphere, and (c) sphere-sphere.

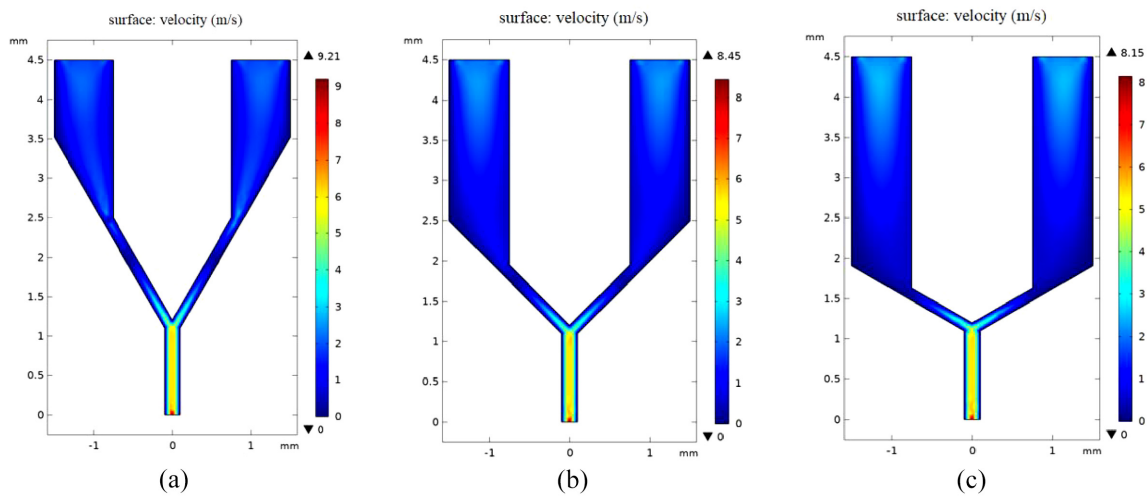


Figure 21. Cloud diagram of glue velocity corresponding to different cone angles: (a) 60° cone angle, (b) 90° cone angle, and (c) 120° cone angle.

outlet increases constantly. At the same time, the inflow of the glue in the nozzle increases with the pressure, which makes the diameter of the dispensing droplet increase.

3.3. Influence of needle displacement on dispensing performance

Similar to the simulation in the previous section, under the condition that other parameters are unchanged, the maximum flow velocity of the glue and the diameter of the glue droplet were solved when the needle displacement is 50, 100, 150, 200, and 250 μm . The result is plotted as the changing curve as shown in Figure 27.

It can be seen from Figure 27 that both the maximum flow velocity of the glue and the diameter of the glue droplets increase with the rise of the driving pressure, and the slope of the change of flow velocity gradually decreases. The reason was analyzed as follows: with the increase of needle displacement, the volume of glue extruded into the nozzle increases, resulting the increase

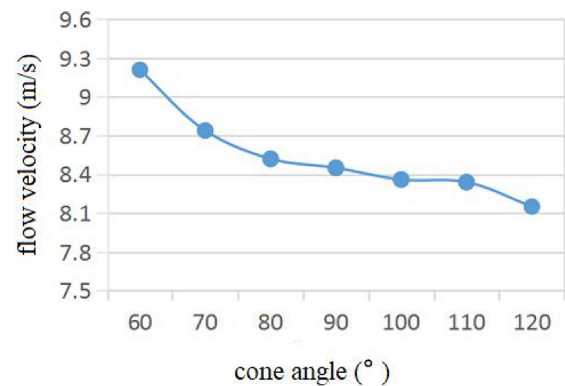


Figure 22. Relation curve between cone angle and flow velocity.

of the diameter of the dispensing droplets. At the same time, the extrusion effect of the needle increases in this process, so that the maximum velocity of the glue in the outlet increases.

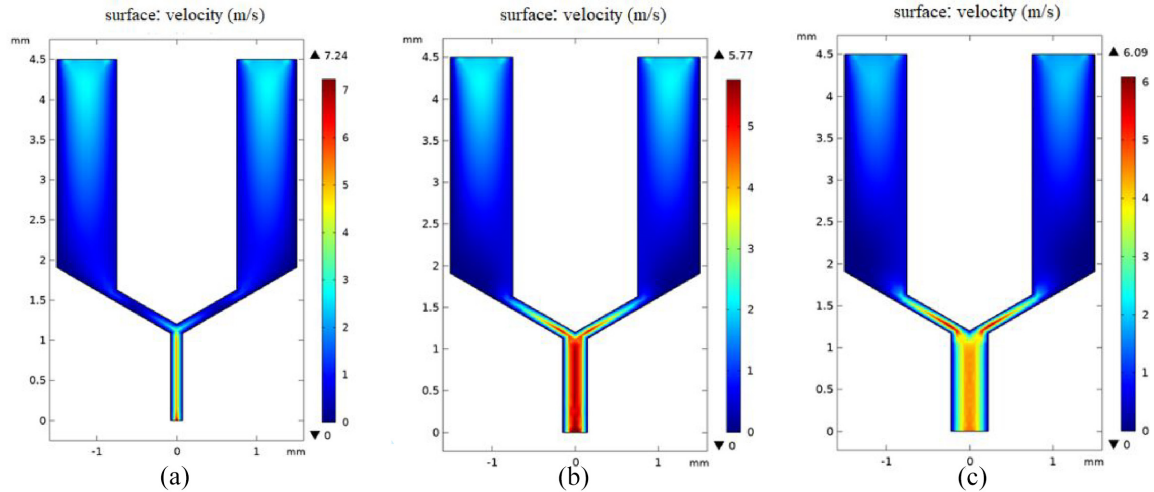


Figure 23. Cloud diagram of glue flow velocity corresponding to different outlet inner diameters: (a) 0.15 mm diameter, (b) 0.3 mm diameter, and (c) 0.45 mm diameter.

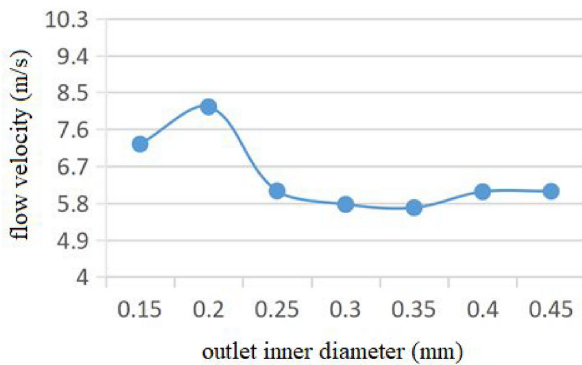


Figure 24. Relation curve of outlet inner diameter and flow velocity.

3.4. Influence of glue viscosity on dispensing performance

When other parameters are the same, the maximum flow velocity of the glue and the diameter of the glue droplet were solved when the glue viscosity was between 200 and 2000 cps. The calculated data points take 200 cps as the interval, and the result is plotted as the changing curve as shown in Figure 28.

As can be seen from Figure 28, the maximum flow velocity of the glue decreases with the increase of its viscosity, and the decrease rate gradually becomes slow. The droplet diameter also decreases slowly with the increase of viscosity. The reason is that with the increase of the viscosity of the glue, the viscosity resistance of the material itself becomes more significant, which makes the flow velocity at the nozzle outlet decrease. At the same time, the fluidity becomes weak, so that the volume of the glue ejecting out of the nozzle reduces, resulting in the decrease of the droplet diameter.

4. Experimental study

4.1. Experimental platform

According to the structure size of the optimized flexible amplifying mechanism, the flexible amplifying mechanism was machined and the piezoelectric jetting dispensing valve was assembled. The diameter of needle is 2 mm and the inner diameter of nozzle outlet is 0.2 mm. The experimental platform was built for experimental study. The experimental platform is mainly composed of the piezoelectric jetting dispensing valve, an air compressor, a function generator (Tektronix AFG1022), a piezoelectric drive controller (HPV-3C0300A0300), a dispensing base plate, etc., as shown in Figure 29. The air compressor is connected with the glue container in the dispensing valve to provide a stable driving pressure. The function generator is connected with the piezoelectric drive controller, which is connected with the piezoelectric actuator (PZT 150/5 × 5/10) to drive the dispensing valve to work. The dispensing base plate is used to collect the experimental glue droplets. Glycerol was mixed with general diluent to produce glue solution, and the viscosity of the glue can be adjusted by changing the concentration of glycerol. In addition, a small amount of dark green colorant was added to the glue to facilitate observation of the droplets.

4.2. Analysis of factors affecting the diameter of glue droplet

In the experiment, the glue droplet is record by a camera. The diameter of the glue droplet can be measured by using the camera imaging principle as equation (13):

$$d_{actual} = \frac{d_{photo} \times S}{f \times a} \quad (13)$$

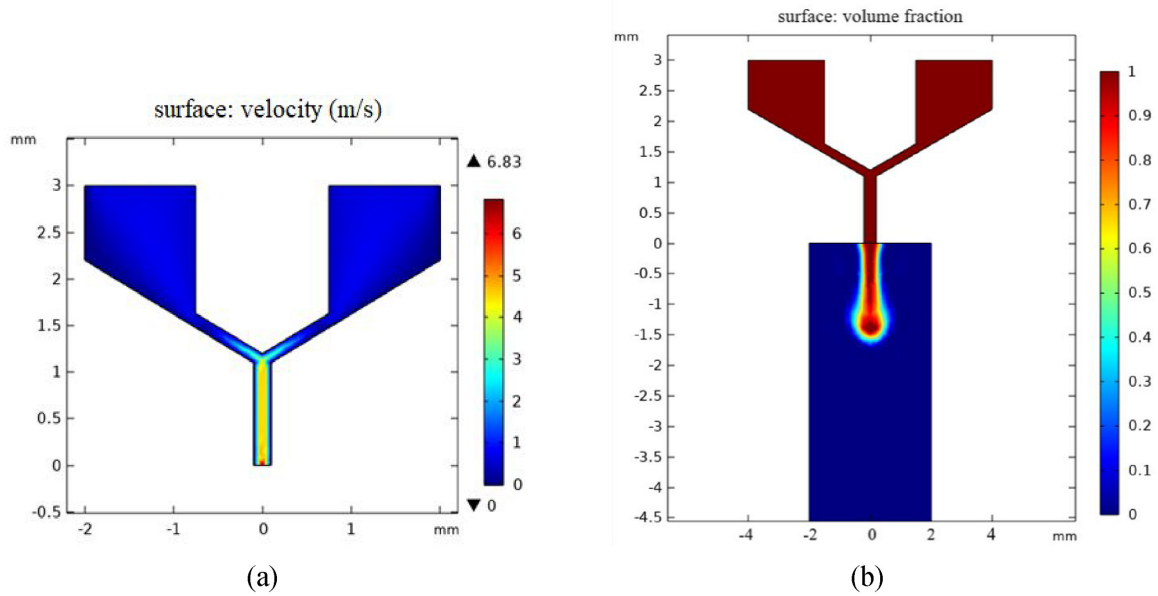


Figure 25. The dispensing performance distribution cloud diagram: (a) velocity of glue and (b) volume fraction of glue.

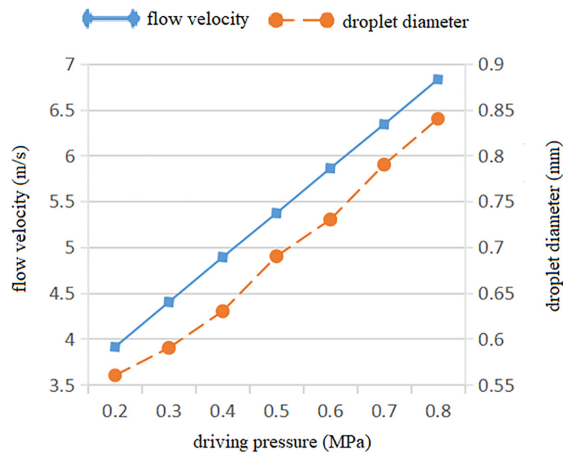


Figure 26. Relation curve of flow velocity, droplet diameter, and driving pressure.

In the equation, d_{photo} is the measured dimension in photograph, S is the distance between the measured object and the lens, f is the focal distance, a is the magnification of the camera sensor to the photograph, which can be obtained by testing larger objects of known dimension and using equation (13) to calculate. So the actual dimension of droplet d_{actual} can be worked out through equation (13) and its d_{photo} is measured in photograph by using a vernier caliper and the accuracy is 0.01 mm.

The diameter was taken as the parameter of the performance analysis to explore the influence law of the driving pressure, the viscosity of the glue, and the driving voltage, which was also compared with the previous simulation results. In order to reduce the experimental error, for the injection results under the same condition,

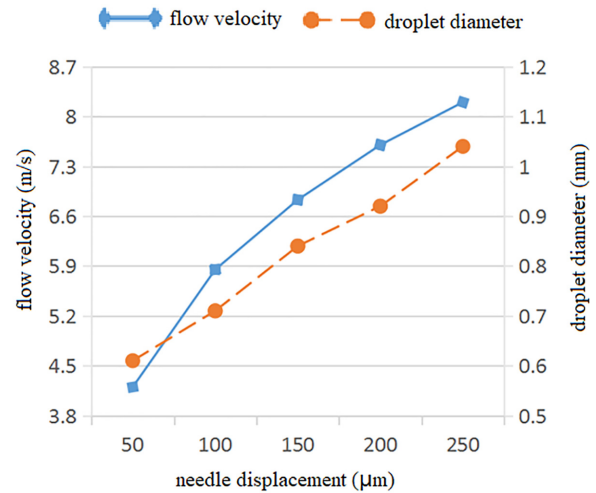


Figure 27. Relation curve of flow velocity, droplet diameter, and needle displacement.

the average diameter of three droplets was taken as the experimental data.

4.2.1. Influence of air pressure on the diameter of glue droplet. It can be seen from Figure 30 that the diameter of the glue droplet increases with the increase of the driving pressure in the experimental study, which has the same influence trend as the simulation analysis. However, when the air pressure is 0.2 MPa, the glue droplet becomes sticky, and cannot eject. It is analyzed that the main reason is that the gas tightness of the device is not enough and the driving pressure is too small, so that the glue does not get enough ejecting momentum. In addition, the diameter of the glue

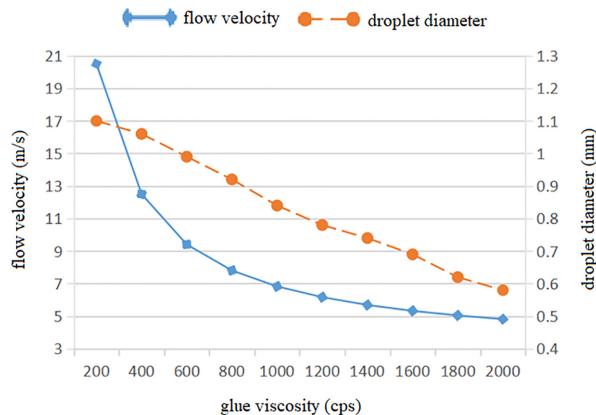


Figure 28. Relation curve of flow velocity, droplet diameter, and glue viscosity.

droplet corresponding to each pressure value in the experiment is larger than the simulation result, because the output displacement of the needle cannot be precisely controlled by controlling the driving voltage of the piezoelectric actuator in the experiment.

4.2.2. Influence of viscosity on the diameter of glue droplet. The corresponding viscosity of the glue was obtained by adjusting the concentration of glycerol. Under the same other conditions, the experimental data of the diameter of the glue droplets were recorded when the viscosity of the glycerol mixture was between 200 and 1200 cps and the step size was 200 cps. After comparison with the simulation analysis results above, the relation diagram is shown in Figure 31.

According to the experimental data curve in Figure 31, the diameter of the glue droplet decreases with the increase of viscosity, and the law is consistent with the

simulation analysis. It was found in the experiment that satellite droplets appeared when the viscosity of glycerol mixture was 200 cps. The main reason is that when the viscosity of the glue is too low, the viscous resistance is too small so that the main droplet is too long, which is easy to form satellite droplets (Chen et al., 2013). In addition, the diameter of the glue droplets corresponding to each viscosity value in the experiment was larger than the simulation result. The main reason is the same as the above experiments, which is caused by the inconsistency between needle displacement and driving voltage of piezoelectric actuator.

4.2.3. Influence of driving voltage on the diameter of glue droplet. The influence law of needle displacement on the diameter of glue droplets was simulated and analyzed above, but the displacement variable cannot be accurately controlled in the experiment. However, the displacement change of needle is actually caused by the driving voltage value of the piezoelectric actuator, so this group of experiments explored the relationship between the driving voltage of the piezoelectric actuator and the diameter of the glue droplets. Under the same other conditions, the experimental data of the diameter of the glue droplets were recorded respectively when the driving voltage was 50–150 V and the step size was 25 V. Some glue droplet samples are shown in Figure 32. The comparison diagram is shown as Figure 33.

It can be seen from Figure 33(b) that the diameter of the glue droplet increases with the rise of the driving voltage, which has the same change trend as Figure 33(a). In addition, as shown in Figure 33(b), the change rate of the droplet diameter gradually decreases with the increase of voltage. The main reason is that the displacement changing curve of the piezoelectric actuator

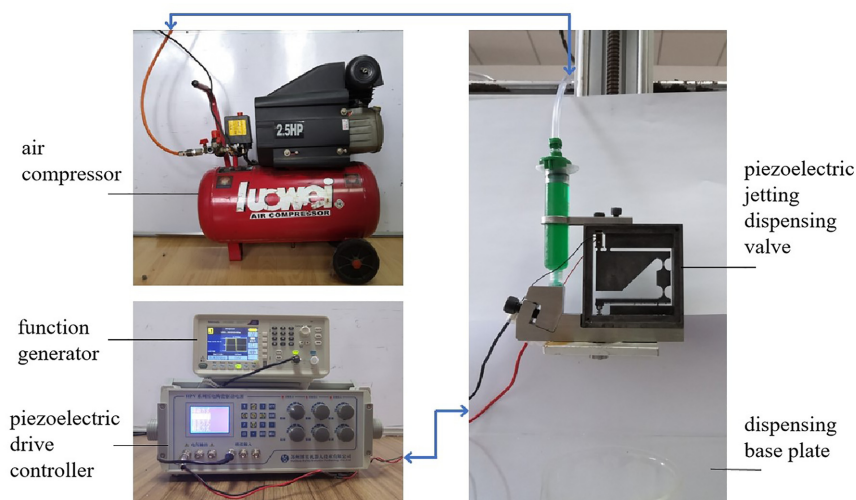


Figure 29. Experimental platform of piezoelectric jetting dispensing valve.

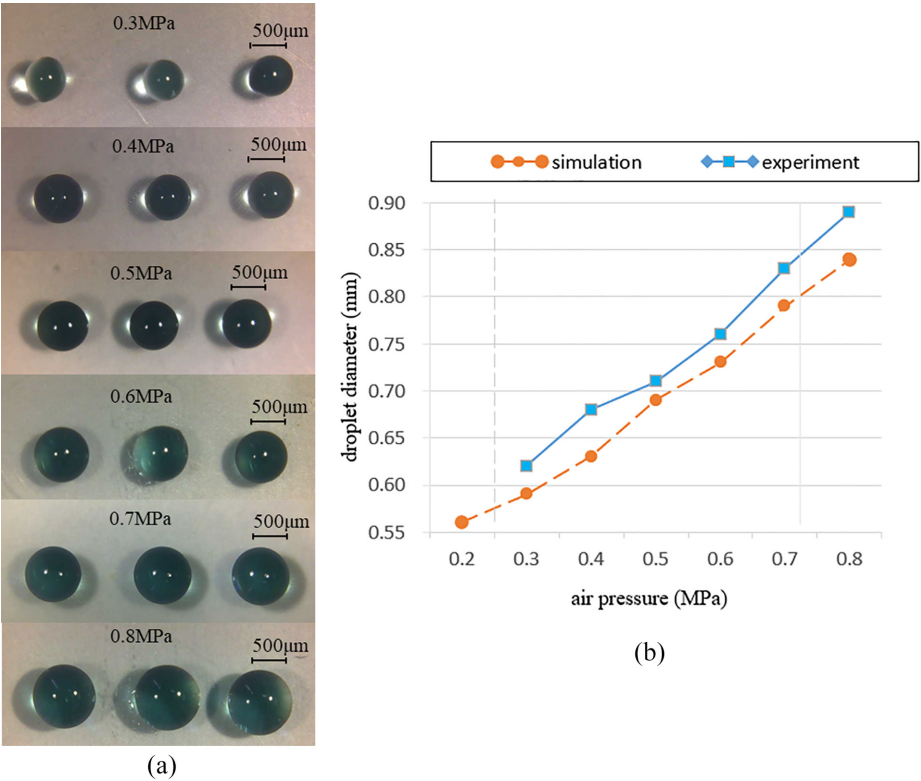


Figure 30. Relationship between air pressure and diameter of glue droplet: (a) samples of droplets and (b) comparison curve of experimental and simulation.

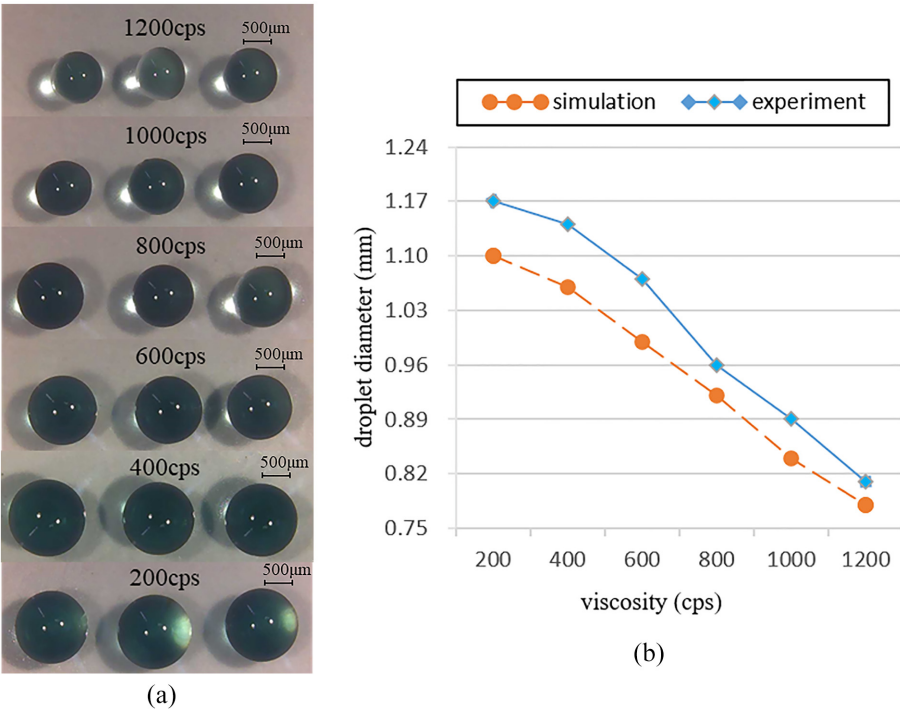


Figure 31. Relationship between viscosity and diameter of glue droplet: (a) samples of droplets and (b) comparison curve of experimental and simulation.

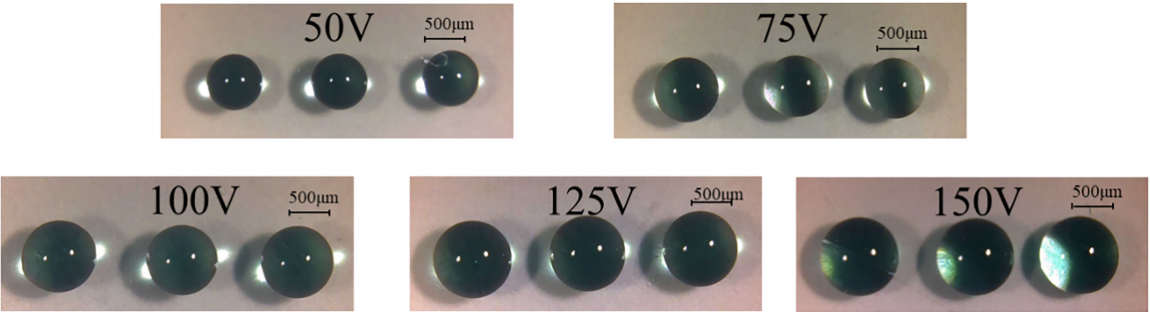


Figure 32. Samples of droplets at different driving voltages.

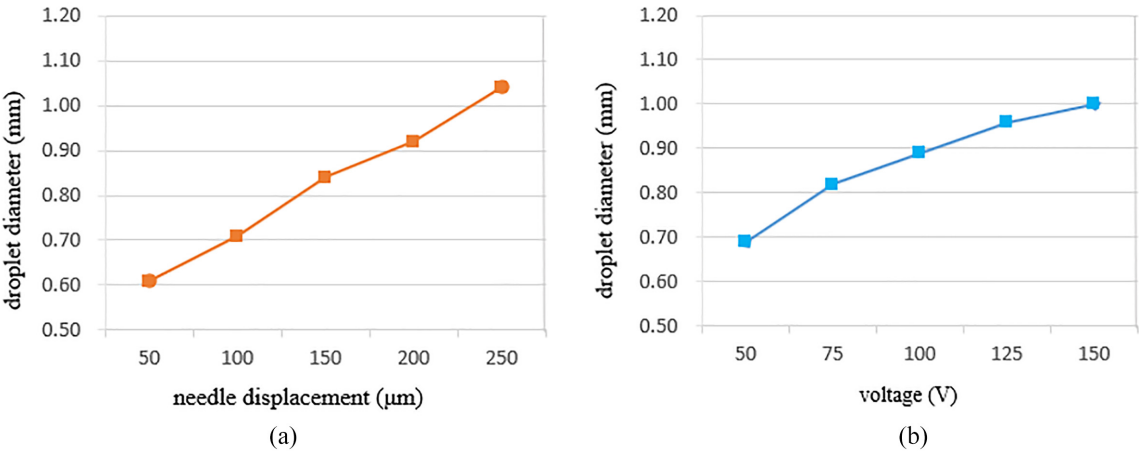


Figure 33. Comparison of the influence of needle displacement and driving voltage on the diameter of glue droplets: (a) needle displacement and (b) voltage.

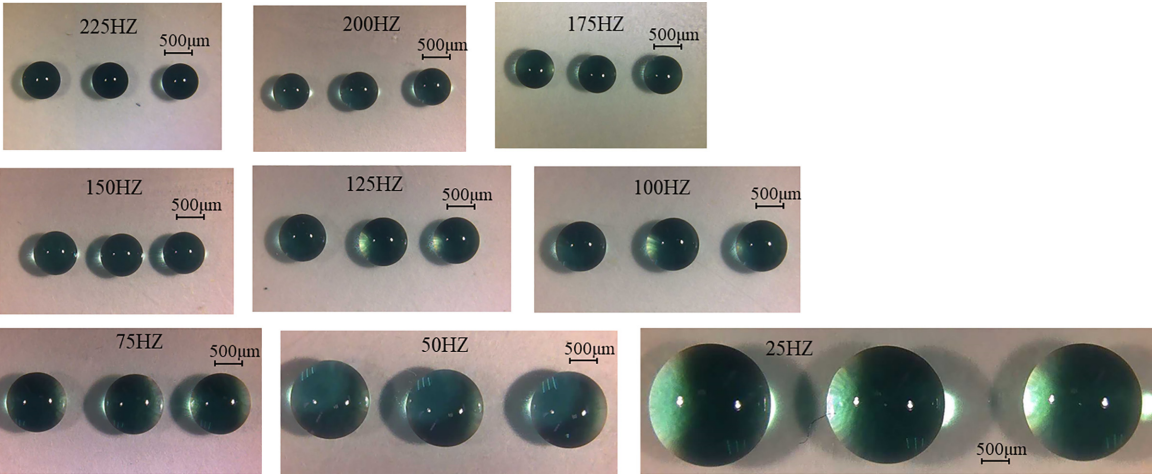


Figure 34. Samples of glue droplets at different frequencies.

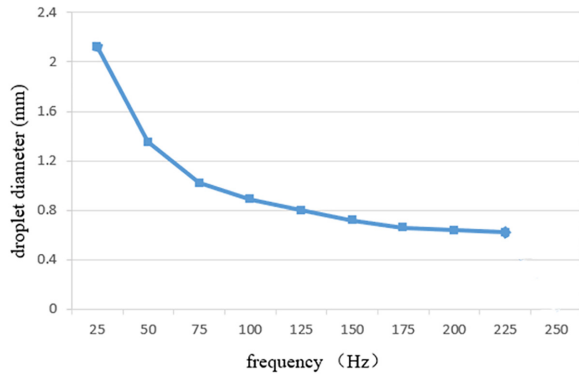


Figure 35. Relation curve between glue droplet diameter and frequency.

Table 2. Experimental data of glue droplet mass.

Mass of glue droplets (mg)									
3.0	3.1	3.0	3.0	3.1	3.3	3.4	3.3	3.0	3.1
3.0	3.4	3.1	3.2	3.3	2.9	3.0	3.3	3.4	2.9
3.0	3.0	3.4	2.9	2.9	3.0	2.9	2.9	2.9	3.1
3.2	3.1	3.3	3.2	3.3	3.1	3.1	2.9	2.9	3.1
3.3	3.4	3.2	3.1	3.1	3.3	3.2	3.3	3.2	3.2

also has a trend of decreasing change rate when the voltage increases.

4.3. Analysis of the maximum dispensing frequency

The working frequency of piezoelectric jetting dispensing valve is an important index to measure its dispensing performance. In general, a higher frequency produces a smaller droplet (Lu et al., 2018). When the voltage frequency gradually increases to a certain frequency, there will be a phenomenon of continuous glue

flow, which cannot create droplets. At that time, the critical voltage frequency is the maximum dispensing frequency. The maximum dispensing frequency can be found by exploring the influence of square wave voltage frequency on the diameter of the glue droplets. The experimental data of the diameter of the glue droplets were recorded when the square wave frequency was 25–250 Hz and the step size was 25 Hz. Some samples of the glue droplets were shown in Figure 34. According to the experimental data, the relation curve between frequency and glue droplet diameter was drawn as shown in Figure 35.

As can be seen from Figure 35, the diameter of the glue droplet decreases with the increase of the square wave frequency, and the rate of change gradually becomes slow. When the frequency reaches 250 Hz, there will be abnormal dispensing phenomenon of continuous glue flow.

In order to explore the more accurate highest dispensing frequency, the dispensing effect was observed at a step of 5 Hz within 225–250 Hz. It was found that relatively even droplets could still be ejected at 230 Hz, but when the frequency was 235 Hz, the jet flow phenomenon would appear. Therefore, the highest dispensing frequency of this dispensing valve prototype is 230 Hz.

4.4. Analysis of dispensing stability

The consistency of the glue droplets reflects the stability of the dispensing valve prototype, and is the main index to analyze the dispensing performance of the dispensing valve prototype. In this group of experiments, the consistency of the dispensing valve prototype was studied according to the mass of each glue droplet. The experiments were carried out under the conditions of driving pressure of 0.8 MPa, glue viscosity of 1000 cps, driving voltage of 100 V, and square wave frequency of 50 Hz.

The glue droplets were collected one by one on the baseplate and put into the electronic balance to record

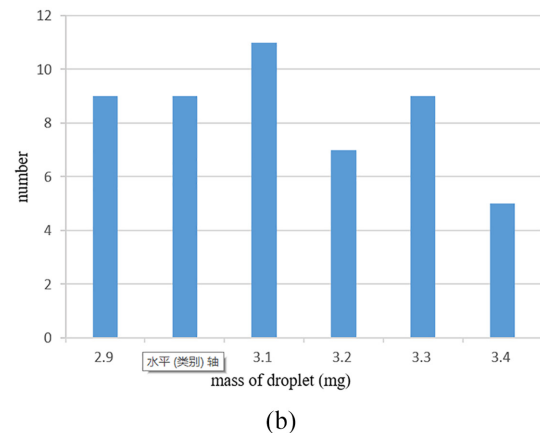
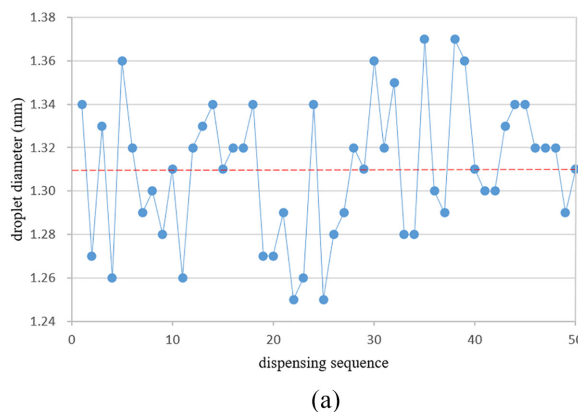


Figure 36. Mass distribution of 50 glue droplets: (a) scatter diagram and (b) bar diagram.

Table 3. Five levels corresponding to the four factors.

	Air pressure (MPa)	Viscosity (cps)	Voltage (V)	Frequency (Hz)
Level 1	0.4	400	100	100
Level 2	0.5	600	110	125
Level 3	0.6	800	120	150
Level 4	0.7	1000	130	175
Level 5	0.8	1200	140	200

the total mass of the baseplate and the glue droplets. After removing the mass of the baseplate, the mass of a single glue droplet was obtained. The data were shown in Table 2 according to the dispensing sequence. According to the experimental data, the scatter diagram and the bar diagram of the glue droplet mass are drawn as Figure 36.

It can be seen from Figure 36 that the mass of glue droplet fluctuates between 2.9 and 3.4 mg with a roughly normal distribution. Among them, the mass of glue droplet with the most concentrated distribution is 3.1 mg, which appears 11 times. It is calculated that the average mass of 50 glue droplets in this group of experiments is 3.126 mg, and the maximum mass deviation is 0.274 mg. So according to equation (14), the

deviation of consistency of glue droplets is about $\pm 8.77\%$.

$$C_m = \pm \frac{|m - \bar{m}|_{\max}}{\bar{m}} \quad (14)$$

4.5. The minimum diameter of dispensing droplet

In the integrated circuit packaging and other fields, piezoelectric jetting dispensing valve is required to have the performance of micro-scale dispensing, so the minimum diameter of the dispensing droplet is also an important index to evaluate the performance of the dispensing valve. Since the four parameters of driving pressure, glue viscosity, driving voltage, and square wave frequency all have an influence on the diameter of the glue droplet, and each parameter has a wide range, the minimum diameter of the glue droplet cannot be obtained by testing all the parameters. Therefore, five levels were set for each parameter and orthogonal experiment was adopted to obtain the best parameter combination. According to the most commonly used value range of each parameter in the actual dispensing production, five level values corresponding to the four factors were set as shown in Table 3. The experiment

Table 4. Table of orthogonal experiment data of glue droplet diameters.

	Pressure	Viscosity	Voltage	Frequency	Diameter (mm)
Experiment 1	Level 1	Level 1	Level 1	Level 1	0.95
Experiment 2	Level 1	Level 2	Level 2	Level 2	0.86
Experiment 3	Level 1	Level 3	Level 3	Level 3	0.76
Experiment 4	Level 1	Level 4	Level 4	Level 4	0.65
Experiment 5	Level 1	Level 5	Level 5	Level 5	0.57
Experiment 6	Level 2	Level 1	Level 2	Level 3	0.91
Experiment 7	Level 2	Level 2	Level 3	Level 4	0.82
Experiment 8	Level 2	Level 3	Level 4	Level 5	0.72
Experiment 9	Level 2	Level 4	Level 5	Level 6	0.87
Experiment 10	Level 2	Level 5	Level 1	Level 2	0.68
Experiment 11	Level 3	Level 1	Level 3	Level 5	0.87
Experiment 12	Level 3	Level 2	Level 4	Level 1	1.02
Experiment 13	Level 3	Level 3	Level 5	Level 2	0.93
Experiment 14	Level 3	Level 4	Level 1	Level 3	0.74
Experiment 15	Level 3	Level 5	Level 2	Level 4	0.62
Experiment 16	Level 4	Level 1	Level 4	Level 2	1.08
Experiment 17	Level 4	Level 2	Level 5	Level 3	0.98
Experiment 18	Level 4	Level 3	Level 1	Level 4	0.80
Experiment 19	Level 4	Level 4	Level 2	Level 5	0.69
Experiment 20	Level 4	Level 5	Level 3	Level 1	0.85
Experiment 21	Level 5	Level 1	Level 5	Level 4	1.03
Experiment 22	Level 5	Level 2	Level 1	Level 5	0.85
Experiment 23	Level 5	Level 3	Level 2	Level 1	1.00
Experiment 24	Level 5	Level 4	Level 3	Level 2	0.91
Experiment 25	Level 5	Level 5	Level 4	Level 3	0.81
Average 1	0.758	0.968	0.804	0.938	/
Average 2	0.800	0.942	0.816	0.892	/
Average 3	0.836	0.842	0.842	0.840	/
Average 4	0.880	0.772	0.856	0.784	/
Average 5	0.920	0.706	0.876	0.740	/
Deviation	0.162	0.262	0.072	0.198	/

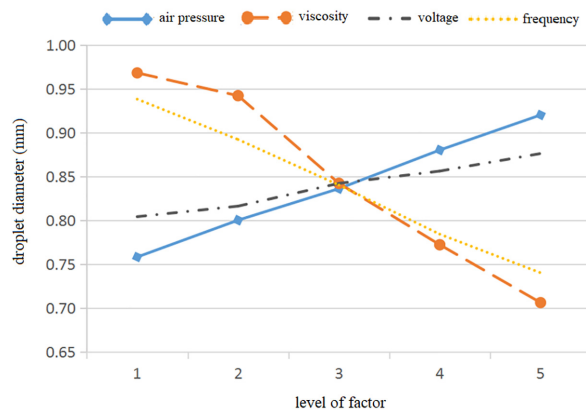


Figure 37. Comparison of each factor affecting the diameter of the droplets.

was arranged according to the orthogonal table L_{25} (5^4), 25 typical parameter groups were set to record diameters of glue droplets respectively. The experimental data is shown in Table 4.

According to the 25 groups of experimental data in Table 4, the average value and deviation of the droplet diameters were calculated when each factor was at the same level. For example, “average1” under column “pressure” represents the average value of five groups of experiment results when the pressure is level 1. And the comparison diagram is shown in Figure 37. It can be concluded that the order of influence of each parameter on the diameter of the glue droplet is: viscosity > frequency > pressure > voltage. Therefore, the best processing parameter combination of minimum glue droplet diameter can be obtained as follows: the driving pressure is set at level 1 (0.4 MPa), the viscosity of glue is set at level 5 (1200 cps), the driving voltage is set at level 1 (100 V), and the square wave frequency is set at level 5 (200 Hz). Under the best processing parameters, the dispensing experiment of the dispensing valve prototype was carried out, and the minimum diameter of the glue droplet was 0.54 mm.

5. Conclusion

In this paper, a piezoelectric jetting dispensing valve for microelectronic packaging is proposed. The valve consists of a piezoelectric actuator and a two-stage displacement flexible amplifying mechanism. The piezoelectric actuator is driven by piezoelectric power supply to generate high-frequency micro-displacement vibration, which is amplified by two-stage displacement amplifying mechanism and make the needle get sufficient displacement. And then the dispensing valve realizes high-frequency micro-scale dispensing through the extrusion action with the nozzle.

According to the displacement requirements of piezoelectric jet dispensing valve, a new type of two-

stage displacement flexible amplification mechanism is proposed, and its structure is designed. The finite element simulation analysis of the stress, strain, and total displacement of the piezoelectric dispensing valve is carried out by ANSYS. Taking the output displacement of the two-stage displacement flexible amplification mechanism as the optimization objective, the structure and size of the flexible amplification mechanism are optimized. Based on the two-stage displacement amplification mechanism, the finite element software COMSOL is used to simulate and analyze the fluid domain in the nozzle and the needle of the piezoelectric jetting dispensing valve. The influences of driving pressure, needle displacement, and viscosity of the glue on the flow velocity of the glue and the diameter of the glue droplet are studied, and the rationality of the design of the valve structure is verified. An experimental platform is built to evaluate the performance of the piezoelectric jetting dispensing valve prototype. The experiment shows that the highest dispensing frequency is 230 Hz, the consistency deviation of dispensing is $\pm 8.77\%$, and the minimum diameter of the glue droplets is 0.54 mm, which meets the need of high frequency and stable micro-scale dispensing.

Declaration of conflicting interests

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Data availability statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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Appendix

Notation

Notation	Meaning
a	Magnification of the camera sensor to the photograph
b_1	Thickness of the middle part at O
b_2	Thickness of the middle part at A
b_3	Thickness of the middle part at B and C
b_4	Thickness of the middle part at D
C_m	Deviation of consistency of glue droplets
d	Nozzle aperture
d_{actual}	Actual dimension of droplet
d_{photo}	Measured dimension of droplet in photograph
F	Focal distance of the camera
F_i	Axial force at i and $i = 1, 2, 3, 4$, and 5 correspond to the five flexible hinges in Figure 5
l_1	Distance from the input end of the first-stage lever to the hinge J_2 shown in Figure 4
l_2	Distance from the output end to the hinge J_2 shown in Figure 4
l_3	Distance from the input end of the second-level lever to the hinge J_5 shown in Figure 4
l_4	Distance from the output end to the hinge J_5 shown in Figure 4
l_i	A series of dimensions parameter of the second optimization shown in Figure 13, $i = 1, 2, 3, 4, 5$, and 6
M_i	Torque at i and $i = 1, 2, 3, 4$, and 5 correspond to the five flexible hinges in Figure 5
J_1	Flexible hinges at O in Figure 3
J_2	Flexible hinges at A in Figure 3
J_3	Flexible hinges at B in Figure 3
J_4	Flexible hinges at D in Figure 3
J_5	Flexible hinges at E in Figure 3
K_α	Rotational stiffness of the flexible hinge
K_x	Tensile stiffness of the flexible hinge
R_1	Radius of the arc at O and A of the flexible hinges in Figure 9
R_2	Radius of the arc at B and C of the flexible hinges in Figure 9
R_3	Radius of the arc at the D flexible hinge in Figure 9
Re	Reynolds number
S	Distance between the measured object and the lens
U	Fluid velocity
U_1	Actual input displacement of the first-stage lever
U_3	Actual output displacement of the first-stage lever
$U_{4'}$	Output displacement of the second-stage lever without considering the compression deformation of hinge J_5
U_4	Actual output displacement of the second-stage lever
U_{in}	Input displacement of the mechanism
U_{out}	Output displacement of the mechanism
W_e	The Weber number
X_1	Dimension parameter of the second optimization shown in Figure 13
X_2	Dimension parameter of the second optimization shown in Figure 13
α_i	Rotation angle at i and $i = 1, 2, 3, 4$, and 5 correspond to the five flexible hinges in Figure 5
Δ_i	Axial deformation at i and $i = 1, 2, 3, 4$, and 5 correspond to the five flexible hinges in Figure 5
Θ_1	Rotation angle of the first lever in Figure 5
Θ_2	Rotation angle of the second lever in Figure 5
M	Hydrodynamic viscosity
ρ	Fluid density
Σ	Surface tension coefficient